Causal Analysis of TB Treatment Abandonment and Retreatment

Synthesis Report & Key Findings

1. Overview & Methodological Challenge

Evaluating the long-term mortality impact of Tuberculosis (TB) treatment abandonment and the subsequent return to care within the São Paulo cohort was heavily confounded by two distinct observational biases: 1. **Immortal Time Bias:** Patients must survive long enough to abandon late in therapy, artificially making late abandonment appear deceptively safe. 2. **Depletion of Susceptibles (Sick-to-Stay):** Patients dying of profound acute TB usually crash in their first month of therapy, leaving no runway to abandon. Because standard treatment mathematically absorbs these inevitable early deaths, abandonment superficially appears protective in crude time-dependent models.

To overcome these unmeasured clinical selections, we utilized a Triangulated Causal Framework—deploying Parametric G-Computation, Marginal Structural Models (IPW), 180-Day Landmark Analyses, and Sequential Target Trial Emulations.

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2. The Mortality Penalty of Abandonment

We definitively proved that abandoning therapy is catastrophic, regardless of timing, effectively destroying the hypothesis that late abandoners are "functionally cured."

Timing of Abandonment (Sequential Target Trials)

To eliminate immortal time bias without a static landmark, we generated six nested Target Trials, repeatedly matching individuals who abandoned exactly in Month *X* exclusively against individuals who were *still actively on treatment* during Month *X*.

Timing of Abandonment	Adjusted Hazard Ratio (aHR)*	95% CI
Month 1	1.84	1.51 - 2.25
Month 2	2.28	1.80 - 2.88
Month 3	2.45	1.94 - 3.10
Month 4	2.26	1.78 - 2.88
Month 5	2.35	1.85 - 2.98
Month 6	2.11	1.66 - 2.68
**Compared to remaining active on standard therapy during that exact month. Adjusted for 13 baseline covariates including HIV, homelessness, and structural poverty.*		

Clinical Takeaway: The relative penalty of abandoning spikes *after* Month 1. By Month 3, compliant patients have structurally stabilized, radically dropping their baseline risk. A patient who arbitrarily stops their antibiotics at Month 3 radically sabotages an otherwise excellent recovery, suffering a massive 2.45-fold relative hazard.

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3. The Paradox of Retreatment (The Failure of Passive Return)

When evaluating patients who abandon, return to the clinic was clinically hypothesized to be a protective, life-saving behavior. Our analyses systematically proved that return to care functions instead as a distress beacon for severe relapse.

A. The Baseline Severity of Retreatment (Marginal Structural Model)

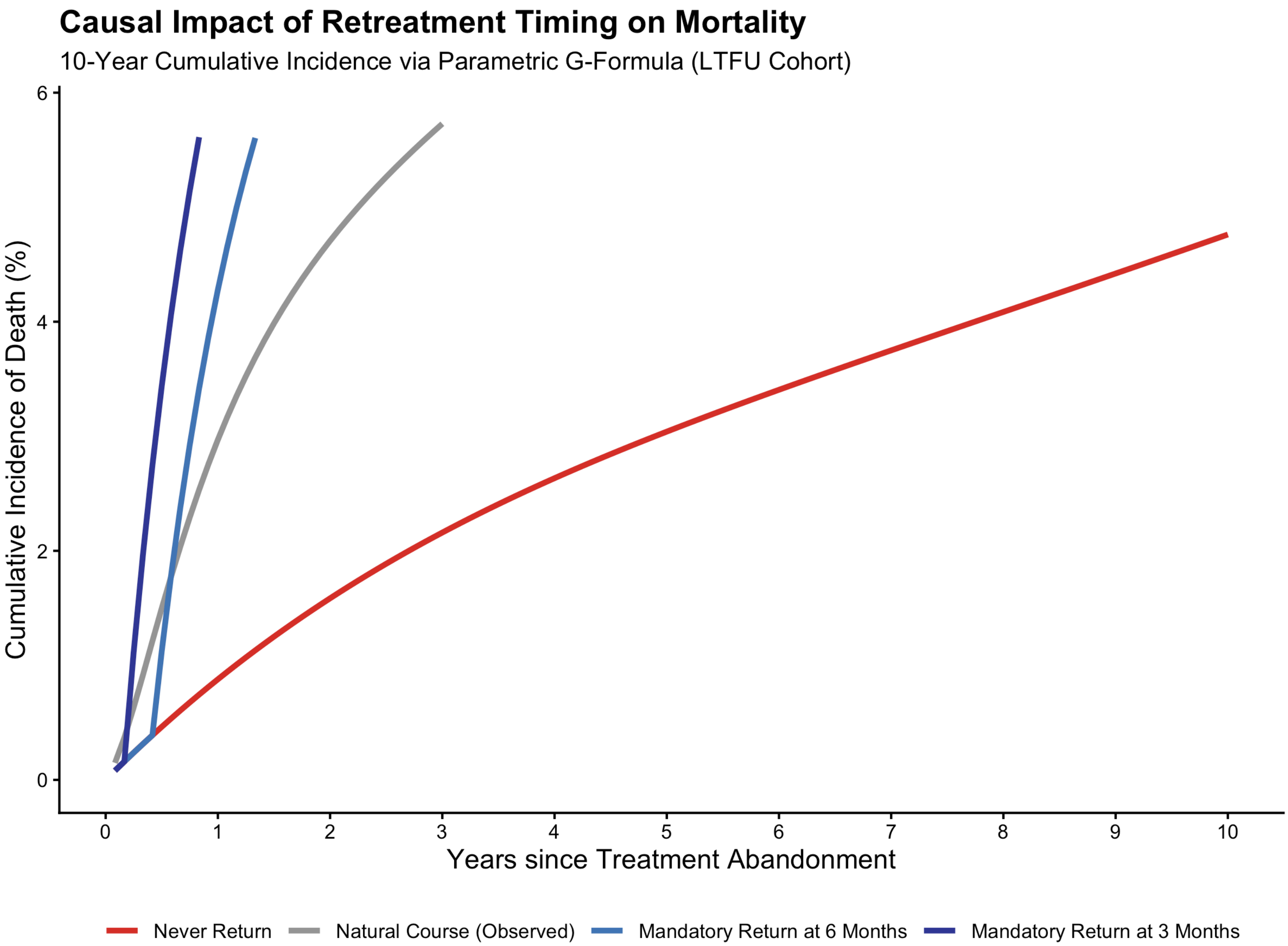
To assess whether returning to therapy actually rescued patients, we utilized an Inverse Probability Weighted Marginal Structural Model (MSM), isolating retreatment as a time-varying exposure. Paradoxically, the MSM revealed that **returning to care is associated with an overwhelming 5-fold increase in mortality (aHR ~5.0)** compared to remaining absent.

Because symptomatic presentation dynamically dictates *who* decides to return, statistical modeling cannot reverse unmeasured confounding by indication. If patients wait to return until they are coughing up blood, no matching algorithm can balance them against patients stabilizing at home. Thus, retreatment acts as a massive **distress beacon for acute, symptomatic relapse**. The current system forces patients to wait until their disease is failing so severely that it physically drives them back to the clinic—at which point their risk is 5x higher.

B. Independence of Gap Duration

Among the subset who returned to therapy, we analyzed whether the length of time "lost to follow-up" predicted mortality, modeling prospective survival directly from the moment of re-entry. - **<2 Month Gap**: Reference (1.00) - **2 - 6 Month Gap**: aHR 1.08 *(p = 0.38)* - **>6 Month Gap**: aHR 1.07 *(p = 0.46)* The physical duration lost to the wind has **zero** effect on outcomes. This proves return to care is gated by a conserved biological threshold: regardless of whether their disease took 1 month or 12 months to progress to severe symptoms, patients re-enter the registry at exactly the same critical state of biological failure.

> [!IMPORTANT] **Simulated Timing of Return (G-Formula)**: The severity bias of retreatment is visually illustrated below via Parametric G-Computation. Simulation reveals that individuals who return to care **Early** (e.g., Month 3) experience dramatically higher cumulative 10-year mortality (**16.2%**) than individuals who **Never Return** (**4.8%**). Returning early is not a sign of "good compliance"; it is a clinical proxy indicating that a patient's individual TB relapse was so aggressive and fast-acting that it biologically forced them back to the ER at Month 3. Conversely, "Never Return" acts as a proxy for spontaneous stabilization, confirming that those who never felt sick enough to return logically suffer the lowest mortality.



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4. Subgroup Effect Modification

We tested for effect modification across the 180-Day Landmark cohort to evaluate who suffers the worst penalty when dropping out.

Subgroup Axis	Demongraphic Stratum	Abandonment aHR	$P_{interaction}$
Age	Age 15-24	4.14	<0.001
	Age 25-44	2.69	
	Age 45-64	1.92	
HIV/AIDS	HIV Negative	2.45	0.027
	HIV Positive	1.63	
Homelessness	Not Homeless	2.50	<0.001

Subgroup Axis	Demongraphic Stratum	Abandonment aHR	$P_{interaction}$
	Homeless	1.37	
Sex	Female	3.19	<0.001
	Male	2.29	

The Competing Risk Paradigm: The mathematical penalty of abandoning treatment is structurally worse for patients who are biologically and socially *healthiest* (e.g., younger individuals, HIV-negative females). A young (15-24) or stable patient has an incredibly low chance of dying while on active TB therapy. If they abandon, they throw away a near-guaranteed cure, skyrocketing their relative likelihood of dying (HR >4.0).

Conversely, marginalized populations (homeless, HIV+) have catastrophically high baseline risks *even when compliant with therapy* due to competing structural pathologies. Furthermore, their integration into social and emergency services makes them highly likely to be recaptured by the health system. Therefore, their absolute time completely "lost" is shorter, and the relative penalty of dropping their TB pills is blunted by their tragically high baseline structural mortality.